Homophily and Differential Returns: Sex Differences in Network Structure and Access in an Advertising Firm

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This paper argues that two network mechanisms operate to create and reinforce gender inequalities in the organizational distribution of power: sex differences in homophily (i.e., tendency to form same-sex network relationships) and in the ability to convert individual attributes and positional resources into network advantages. These arguments were tested in a network analytic study of men's and women's interaction patterns in an advertising firm. Men were more likely to form homophilous ties across multiple networks and to have stronger homophilous ties, while women evidenced a differentiated network pattern in which they obtained social support and friendship from women and instrumental access through network ties to men. Although centrality in organization-wide networks did not vary by sex once controls were instituted, relative to women, men appeared to reap greater network returns from similar individual and positional resources, as well as from homophilous relationships.•

Ample research has documented the sex structuring of organizations, including the segregation of men and women into different jobs, occupations, firms, and industries (Baron and Bielby, 1985, 1986; Bielby and Baron, 1986), patterns of selective recruitment and advancement (Acker and Van Houten, 1976; Baron, Davis-Blake, and Bielby, 1986), and salary differentials (Treiman and Hartman, 1981; Drazin and Auster, 1987). Yet despite voluminous anecdotal and survey research indicating that women in organizational settings lack access to or are excluded from emergent interaction networks (e.g., Kanter, 1977; Harlan and Weiss, 1982; Ragins and Sundstrom, 1989; O'Leary and Ickovics, 1992), little empirical research has investigated the role of networks in creating or reinforcing gender inequalities. Because the differential allocation of network rewards may partially account for gender differences in career outcomes (Brass, 1985; Morrison and Von Glinow, 1990; Smith-Lovin and McPherson, 1993), the role of sex differences in informal networks warrants further investigation.

Most studies reporting gender differences in workplace relational patterns, with few exceptions, have not systematically examined the interaction networks in question, relying instead on anecdotal accounts of perceived exclusion. Network analysis, however, explores actual relationships that evolve among individuals rather than their feelings about and perceptions of their social involvement (Miller, 1975) or their membership in categories used as proxies for likely interaction patterns (Wellman, 1983). Further, any set of dyadic relationships is embedded in a structural context that allocates greater privilege and power to certain actors and that, therefore, must be taken into account in any empirical exploration (Krackhardt, 1990). Only network analysis can uncover systematic differences in the ways in which men and women are located in an organizational context shaped by both formally prescribed and emergent patterns of interaction.

The scant network analytic research on women in organizations (Miller, Labovitz, and Fry, 1975; Lincoln and

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Miller, 1979; Miller, Lincoln, and Olson, 1981; Olson and Miller, 1983; Brass, 1985; Miller, 1986) provides more systematic evidence of informal barriers faced by women in organizations. This body of research, however, has lacked well-developed theoretical explanations for differences in network access and has not clearly specified relevant network types. Further, the mechanisms through which network rewards may be differentially allocated to men and women have not been sufficiently examined. Overall, additional empirical evidence and theoretical development are needed to clarify the ways and extent to which men's and women's networks differ, as well as potential consequences of observed differences.

This paper explores sex differences in network structure and access in five interaction networks within one organization. To develop a better understanding of whether and how reported disadvantages to women are realized, I explored gender differences at two levels of network analysis—properties of individuals' sociometric choices and indices of centrality within organization-wide networks—as well as differences in the extent to which formal positions and individual attributes contribute to network access.

ACCESS TO INTERACTION NETWORKS: THEORY AND HYPOTHESES

A variety of theoretical perspectives have been offered to explain or predict sex differences in network structure and access. Person-centered or dispositional explanations (Riger and Galligan, 1980; Downey and Lahey, 1988) focus on individual preferences, personality characteristics, and behavior patterns that differentiate men from women. Situation-centered or structural explanations, by contrast, argue that characteristics of individuals' social contexts, rather than their traits, account for the principal obstacles women encounter in the workplace (Riger and Galligan, 1980; Downey and Lahey, 1988; Moore, 1990; Smith-Lovin and McPherson, 1993). What has been missing in the literature are systematic attempts to submit competing perspectives to empirical test and integrate and extend those that are complementary. For example, explanations based on the notion of preference for homophily, which exemplifies the individual perspective, can be tested and extended by explicitly taking into account structural constraints on preferences.

Homophily in Network Relationships

A widely cited explanation for women's purported exclusion or limited access to interaction networks is preference for homophily, i.e., interaction with others who are similar on given attributes such as sex, race, and education (Rogers and Kincaid, 1981). Similarity of personal characteristics implies common interests and worldviews and best explains the formation of expressive ties based on interpersonal attraction (Lazersfeld and Merton, 1954; McPherson and Smith-Lovin, 1987; Marsden, 1988). Social homogeneity in the workplace makes communication easier, behavior more predictable and fosters relationships of trust and reciprocity

(Lincoln and Miller, 1979), thus also enhancing instrumental relationships.

Although homophily need not be defined exclusively in terms of gender, following Kanter's (1977) description of the formation of closed social circles that limit women's participation in workplace informal networks, numerous studies have indicated the prevalence of sex-based homophily in workplace relationships and its detrimental consequences for women. Brass (1985), in a study of a gender-balanced nonsupervisory group, observed that although both sexes were equally central to interaction networks, men and women tended to interact within sex-segregated networks. Moreover, women in his study firm, with the exception of those who were part of mixed-gender workgroups, were less central to those networks associated with future promotion: men's networks and the interaction network of the "dominant coalition." South et al. (1982) found that employees in a large federal bureaucracy received the most social support from coworkers of the same sex. Tsui and O'Reilly (1989) noted that superior-subordinate dissimilarity is associated with lower effectiveness, as perceived by superiors, and superiors feeling less interpersonal attraction or liking for the subordinates, with particularly strong effects in mixed-gender dyads.

Two theoretical perspectives on homophily have been offered. The first derives from theories of interpersonal attraction, focusing on individual preferences for relationships with similar others, to explain the sex-segregation of interaction networks (Lincoln and Miller, 1979; Brass, 1985; Marsden, 1988). The alternate, structural perspective argues that availability for contact precedes and limits individual preferences, since the extent of contact between male and female coworkers will be constrained by the sex composition of the group (Blau, 1977). McPherson and Smith-Lovin (1987) have suggested distinguishing between induced homophily, which is the result of availability constraints, and choice homophily, resulting from individual preferences.

Finding support for the preference perspective, therefore, requires controlling for availability, since what may be interpreted as a male preference for homophily may be strictly due to a higher prevalence of males within a given firm. Homophily theories, however, focus exclusively on preferences for similarity and are therefore limited as explanations for instrumental action. In the workplace, where social familiarity may not always be the principal consideration, individuals are also likely to "prefer" interaction with higher-status others in order to gain access to valued resources (Lin, 1982). But demographic similarity and status are not independent: Ascribed attributes such as race and gender are often highly correlated with organizational positions, such that homophily on one dimension often implies homophily on others (Blau, 1977; Smith-Lovin and McPherson, 1993). Thus, if one gender group possesses greater status and resources, its members are likely to evidence a greater preference for within-group interaction, while members of the lower-status group may prefer intergroup interaction in order to gain access to resources. The key point is that preferences for homophily and status will tend to coincide for men and exist in competition for women. This, in and of itself, constitutes a form of structural constraint, making it difficult to attribute network choices to individual factors such as preference.

One potential response to such constraints is for women to seek different types of network resources from different sources. To be effective in their jobs without forfeiting access to the social benefits associated with expressive work relationships, women may attempt a network "division of labor," obtaining social support and friendship from ties to female coworkers but relying on relationships with male coworkers to gain access to instrumental resources (Lipman-Blumen, 1980). It has been argued that such heterophilous ties are more likely to provide support for the advancement of women when women are in the minority (South et al., 1987; Aldrich, 1989). Thus, whether due to interaction preferences or strategies for access to resources, men are able to obtain both expressive and instrumental network resources from male colleagues, while women are likely to navigate a course between two different social circles, choosing women as expressive network contacts but relying on men for their instrumental linkages, which leads to the following hypothesis:

Hypothesis 1: Controlling for availability, women will tend to choose women as expressive network contacts but will choose men as instrumental network contacts. Men will tend predominantly to choose men across multiple networks.

While seemingly adaptive, such a network differentiation may have hidden costs or unintended consequences for women. An important factor is multiplexity, which refers to the number of exchange contents in any relationship (Barnes, 1972; Lincoln, 1982), for example, whether an advice relationship is also a friendship relationship. The greater the multiplexity of a relationship, the stronger the link (Tichy, 1981; Brass, 1992), which, in turn, is associated with the extent to which the relationship is stable, reciprocal, and intimate (Granovetter, 1973; Lincoln, 1982). Given that sentiments and influence are more likely to be mobilized by strong ties or social relations that are close and intense (Stinchcomb, 1987; Krackhardt, 1992), a lower incidence of strong, multiplex relationships may be a significant form of female network disadvantage:

Hypothesis 2: Men will have more multiplex network ties, in total, than women, as well as more multiplex ties to men than women have to women.

Second, if homophily and status of contacts are positively correlated for men and negatively correlated for women, homophily may partially account for sex differences in access to organization-wide networks. Although it has been argued that network centrality is increased by weak ties providing access beyond an actor's immediate social circle (Granovetter, 1973), the advantage provided by weak ties may be contingent on the extent to which they constitute a bridge between individuals or groups possessing scarce or valued resources (Granovetter, 1982; Brass, 1992). Thus, depending on the distribution of power among men and women in a social system, heterophily may provide greater advantage to women than to men. Conversely, homophily may serve to decrease women's access to organization-wide networks while having little effect on men's. Few studies relating gender to network access, however, have explicitly considered the relationships between homophily and centrality. Since there has been little empirical work in this domain, no specific hypotheses are formulated.

Human Capital and Systemic Barriers

A second area of research and theorizing directly explores sex differences in network centrality or degrees of access to and integration within organization-wide interaction webs. Within this realm, the competing hypotheses derive from human capital theory and theories of systemic barriers or institutionalized bias (Miller, 1986). The broader debate concerns the extent to which social rewards (e.g., network access) are allocated on the basis of universalistic, organizationally relevant criteria or, instead, on the basis of particularistic standards such as social familiarity (Pfeffer, Salancik, and Leblebici, 1976) or bias (Miller, 1986).

The human capital perspective (Blau and Ferber, 1987), which has been extended in network research to include a broader set of "bureaucratic investments" (Miller, 1986), argues that differences between men and women with respect to achievement (i.e., education, experience, and expertise) and formal position (i.e., rank, department, occupation) make women, as a group, less desirable network contacts, accounting for most observed disadvantages to women. Sex differences in centrality, therefore, would be due to differences in education, expertise, and positional resources (Miller, Labovitz, and Fry, 1975). Moore (1990), who reported that most gender differences in network composition disappear or are considerably reduced when structural variables such as employment are controlled, found support for this perspective. Similarly, Aldrich (1989) found few gender differences in network structure among established male and female entrepreneurs.

An alternate set of theories also considers sex differences in bureaucratic investments but emphasizes their effects on patterns of interaction and interaction dynamics between majority and minority group members (Kanter, 1977; Lincoln, 1982; South et al., 1982; Alderfer, 1986; McPherson and Smith-Lovin, 1987). In contrast to human capital theories, the "systemic barriers" perspective (Morrison and Von Glinow, 1990) suggests that numerical differences and stratification patterns are amplified through emergent processes that augment the numerically derived power of majority groups (Fairhurst and Snavely, 1983). Thus, from this viewpoint, sex differences in network access reflect the expectations and biases of organizational members (Miller, 1986) and therefore would be expected to persist despite controls for availability, human capital, and formal positions.

In intraorganizational research on centrality both perspectives have received some measure of empirical support. Miller, Labovitz, and Fry (1975), in a study of five professional organizations, found that men received more friendship and communication choices than women and were several times more likely to be cited as being professionally respected and influential. Further, in stark contrast to the predictions of the human capital perspective, these disadvantages were more pronounced for women with advanced education, high rank, and superior authority. Other studies (Miller, Lincoln, and Olson, 1981; Olson and Miller, 1983; Miller, 1986), however, suggest that the direct effects of gender are limited once controls for personal resources and formal positions are instituted. In Brass's (1985) study, for example, the reduced centrality of women in networks associated with future promotion was not observed for women members of integrated work groups.

Given these contradictory findings, the present hypotheses were formulated on the basis of the well-established finding that high-status individuals have more extensive network connections (Lincoln and Miller, 1979; Miller, 1986). Since in many organizations, the present research site included, women have yet to reach the highest management levels in significant numbers and are still disproportionally found in lower-level jobs or in lower-status groups lacking control of critical contingencies, women as a group are hypothesized to be less central. Moreover, since network centrality is increased by connections to others who are highly central, to the extent that men's network relationships are characterized by a high degree of homophily, most network ties among high-status actors will be ties among men. Thus,

Hypothesis 3: Men will hold more central network positions than women in workplace interaction networks.

The findings of Miller, Labovitz, and Fry (1975) also suggest that sex differences in network centrality may be more marked with respect to instrumental networks. This is also consistent with expectation-states theory (Berger, Conner, and Fisek, 1974; Berger et al., 1977); if gender biases are such that both men and women attribute greater task-related competencies to men, men are more likely than women to be sought out as instrumental network contacts. Extrapolating on the research and theory discussed above:

Hypothesis 4: Sex differences in centrality will be higher in instrumental networks than in expressive networks.

If differences between men and women in network centrality are expected, the next logical question is what produces centrality. The variables most often reported in the literature as highly associated with network centrality are hierarchical level (Lincoln and Miller, 1979; Roberts and O'Reilly, 1979; Miller, 1986), seniority (Fombrun, 1983), education and expertise (Lincoln and Miller, 1979; Roberts and O'Reilly, 1979; Miller, 1986), and control of critical organizational contingencies (Miller, 1986).

In this domain, a human capital perspective would predict that men and women receive equivalent network returns for their resources and investments, while a systemic barriers approach would expect greater returns to men. Empirical evidence appears to favor the latter explanation, suggesting that investment yields higher returns for men than for women (see Morrison and Von Glinow, 1990, for a review). While most of the literature has focused on salary and advancement, some empirical evidence suggests that men and women get differential returns with respect to network access. Miller (1986; Miller, Lincoln, and Olson, 1981) found that in contexts in which men and women enjoy equal access to informal networks, predictors of centrality, or ways of gaining access to these networks, varied by sex:

> The informal organizational advantages [men] came to enjoy were a function of their bureaucratic investments, their hierarchical positions, and their work-related resources. The same could not be said of white women . . . [their access was] more likely to be linked to the nature of the clients that they handled, their special ties to the surrounding community outside their programs, or their personal, non-bureaucratic characteristics—age in particular. (Miller, 1986:6)

Consistent findings were reported by Olson and Miller (1983): In a study of a textile-industry firm, males were found to be at an advantage in the ability to convert formal authority into network centrality. Thus,

Hypothesis 5: Men will receive greater network returns on their individual and positional resources than women.

METHOD

Research Site

This research was part of a larger study of informal networks in a New England advertising and public relations agency. The agency was founded in the early 1970s and was privately owned by the top management team. Firm size, as measured by number of employees and billings (advertising revenues) had nearly doubled in the three years prior to this study. This period had been a turbulent one: To maintain net growth the firm had to replace business lost as a result of major client defections and substantial advertising budget cuts by its remaining clients. At the time of data collection, the firm had 94 full-time employees.

Like most advertising firms, the agency was organized into Account Services, Creative Services, Operations (Production and Traffic), Media, and various support departments, including Accounting and Personnel. In this firm, these departments clustered into three larger groups comprising departments that held similar positions in the workflow, were afforded the same level of status, and were located on the same floor of the facility.

At the input boundary, the Account Services and Public Relations (PR) departments sought and developed relationships with clients, gathered the information necessary for the development of an advertising or PR campaign, and provided account management. As in most agencies, these groups were the elite of the firm because they controlled the agency's only source of income—the clients. Their elite status and concomitant power were enhanced by the many uncertainties that characterize the client relationship: In advertising there is typically no specified period of time during which agencies are assured of employment, and it is not unusual for clients to terminate relationships with agencies abruptly (Comanor, Kover, and Smiley, 1981). The status of these two groups was highlighted by their location on the top (third) floor of the facility.

The core of the firm was the Creative, Research, and Production departments, which use the information provided by Account Services and Public Relations to create advertising, test it with a target audience, and produce a print or broadcast ad. These groups worked closely together in the development of a "job" and were housed on the second floor of the firm. Since the national reputation of an agency is contingent on its ability to win prestigious advertising awards, these groups also possessed considerable status.

As in most advertising firms, friction between the creative and client management groups was common. The primary allegiance of Account Services is to the agency, and its job is to please its customers; the Creative staff, however, judges its performance against that of its peers in the profession and is primarily interested in opportunities for unrestricted creativity. Thus, Account Services staff typically viewed the Creative staff as unwilling to consider its client's needs, while the Creative staff argued that Account Services lacked regard for feasibility or creative quality.

On the output boundary, the Media Department places the finished advertisement in newspapers, magazines, and television. The administrative functions included Personnel and Accounting. In contrast to the rest of the firm, the work of all but the top managers of these groups was highly routine and structured. Members of these groups worked on the first floor of the building, in relative isolation from the rest of the firm, and tended to be viewed as "paper pushers," fulfilling no essential function. Both Account Services and the Creative staff were often in dispute with the first-floor support groups, blaming them for the imposition of "bureaucratic" requirements.

Although the firm prided itself in having no formally published organization chart, senior management identified five distinct hierarchical levels: senior management, middle management, nonsupervisory professional, entry-level staff, and secretarial/administrative support staff. Senior management included the president and vice presidents of the agency; those who reported directly to them were middle management. The middle management group also included all the firm's "star" artists and writers who were not vice presidents. The third layer of the firm consisted of nonsupervisory professionals. This strata differed from the next, the entry-level employees, in years of experience, tenure in the firm and job responsibilities. The lowest level, the secretarial/administrative staff, consisted of secretaries, receptionists, and a group of employees called "Account Assistants" (AAs), who performed administrative support functions for the Account Services and Public Relations staff. The AA role was being phased out at the time of data collection, for two reasons: The firm had experienced difficulty promoting members of this group, all female, to the next level, and their job responsibilities overlapped highly with those of the secretarial staff.

Participants

Network analysis requires that data be collected from all members of a system because, at present, no generally accepted techniques have been developed for sampling within a network (Rogers and Kincaid, 1981). This, in turn, requires accurate definition of the boundaries of the relevant "system" or network (Laumann, Marsden, and Prensky, 1983). Laumann, Marsden, and Prensky (1983: 21) proposed two alternate approaches: Using a realist approach entails adopting the point of view of the actors themselves in defining the boundaries of a social entity; using a nominalist approach, the investigator "imposes a conceptual framework to serve his own analytic purposes."

I used a realist approach, drawing inclusion rules on the basis of employees' clearly drawn distinction between the professional and secretarial/clerical employees and limits imposed by management with respect to the number of employees permitted to participate in the study. Thus, all professional staff (N = 73) were included a priori. As a safeguard against bias in the definition of the network boundaries, however, the AA group was also included, and an iterative method was then used to determine which, if any members of the secretarial staff should be viewed as part of the interaction networks studied. This consisted of examining the sociometric choices of the initial respondent group to identify the frequency with which secretarial staff were cited. I made a decision rule to enlarge the network population to include any member of the secretarial staff who was nominated by more than two members of the initial group in answer to the sociometric questions described below. Anyone who was nominated by two or fewer of the initial 73 respondents was deemed to be marginal to the network and was excluded from the study. This method yielded an additional seven respondents for inclusion, whose own sociometric choices, in turn, suggested no further participants to be included in the study. The final response rate was 97.5 percent of the network population, which included 98.6 percent (73 of 74 members) of the originally identified professional population and six of the seven members of the secretarial/clerical staff identified by the iterative method.

Table 1 describes the hierarchical and functional distribution of the 34 men and 45 women in the final respondent group. Overall, men and women appeared to be grouped into different departments and/or hierarchical levels within departments. Women were underrepresented in upper management and in the core departments (Creative, Production, and Research) and overrepresented in the two lowest levels of the organization, as well as in the lower-status departments (Media, Accounting, Personnel). Although the gender composition of the client management groups was more evenly distributed, 44 percent of the women in this group (in contrast to none of the men) belonged to the two lowest hierarchical levels. In this organization, women were predominantly found in either low-status "female" departments or in the lower ranks of the traditionally male groups.

Table 1

Functional and Hierarchical Distribution of Men and Women*

Hierarchical level	Department 1 Support	Department 2 Core	Department 3 Client contact	Subtotals	Total
1 (Secretarial)	2/0†	3/1	4/0	9/1	10
2 (Entry)	10/0	3/0	3/0	16/0	16
3 (Professional)	4/1	2/6	4/4	10/11	21
4 (Middle management)	4/1	1/5	4/7	9/13	22
5 (Senior management)	0/3	0/3	1/3	1/10	10
Subtotals	20/5	9/15	16/14	45/34	
Total	25	24	30		79

* Thirteen of the remaining 14 employees were all female, all belonged to Level 1 (secretarial) and were evenly distributed across departments. The fourteenth was a male middle manager in Department 1.
† The first number refers to the number of females, the second number to the number of males.

Measures

Respondents filled out sociometric and background questionnaires in the course of one-on-one structured interviews with the researcher. In all cases, participation was voluntary, and respondents were assured that their responses would be kept confidential.

Network indices. On the sociometric questionnaire, respondents were asked to name the people in the firm (1) "with whom you discuss what is going on in the organization," (2) who are "important sources of professional advice, whom you approach if you have a work-related problem or when you want advice on a decision you have to make," (3) "that you know you can count on, whom you view as allies, who are dependable in times of crisis," (4) "that you have personally talked to over the past couple of years when you wanted to affect the outcome of an important decision," and (5) "who are very good friends of yours, people whom you see socially outside of work." The caveat "whom you see socially outside of work" was included as part of the friendship question in order to ensure consistency across respondents in the definition of friendship. It is important to note, however, that this wording may contribute to friendship homophily by focusing on relationships that might be perceived by others as romantic, sexual, or nonprofessional.

In an effort to limit measurement error (Holland and Leinhart, 1973), respondents were not restricted to a fixed number of nominations. Although ten blanks were provided after each question, respondents were told that they could add additional blanks as needed, and several individuals did create additional blanks. As an aid to recall, respondents were also provided with the firm's one-page telephone directory listing all members.

The answers to these questions provided the raw data used to define communication, advice, support, influence, and friendship networks. These content domains were chosen to provide a broad spectrum of network transactions and to address the common critique that network studies do not include a wide enough range of network types (Fombrun, 1982; Smith-Lovin and McPherson, 1993). Based on previous conceptualizations (Tichy, Tushman, and Fombrun, 1974; Fombrun, 1982; Krackhardt, 1990), it was presumed that the influence, advice, and communication networks, on the one hand, and the friendship network, on the other, represent instrumental and expressive relations, respectively.

Centrality. Centrality was operationalized as "aggregate prominence" (Knoke and Burt, 1983), which indexes individual centrality as a function of the centrality of those to whom one is connected through direct and indirect links (Bonacich, 1987). Rather than allowing all relationships of equal proximity to contribute equally to an actor's centrality, as in closeness centrality indices (Freeman, 1978), this formulation assumes that centrality is increased positively by connections to others who are highly central and assigns the highest level of centrality to the actors with the closest relations (that is, direct or short indirect links) with many central actors (Bonacich, 1987).[1] Aggregate prominence is calculated as follows (Burt, 1987):

$$g\,C_j = \sum_{i=1}^{N} C_i z_{ij},$$

where C_j = the centrality of actor j, C_i = the centrality of person i, N = the population size, z_{ij} = the distance between actors i and j, which varies from 0 to 1 as the number of indirect links between them decreases, and g = the maximum eigenvalue of the aggregate relation matrix. Centrality scores range from 0 to 1, higher values indicating greater centrality.

In computing the centrality scores, relationships were not symmetrized, thus preserving the distinction between being the source versus being the object of a relation (Burt, 1982), for two reasons. First, many relationships, for example, advice and influence relations, are inherently asymmetrical (Krackhardt, 1990). Even intimate relationships such as friendship ties, however, are often asymmetrical, since conceptions of closeness vary across individuals (Krackhardt, 1990; Marsden, 1990). This was confirmed in analyses of reciprocation rates, which were 37, 32, 47, 19, and 46 percent, respectively, for the communication, advice, support, influence, and friendship networks.[2] Second, it is possible that arbitrary symmetrization of relationships would obscure sex differences in network access or reciprocation, since, as noted by Dexter (1985), if women possess less capital, reciprocity is more difficult to establish. In this organization, women's ties were significantly less likely to be reciprocated than men's in the advice ($p < .001$), support ($p < .05$), and influence ($p < .01$) networks, although these differences disappeared when individual and formal positions were held constant.

Homophily. To operationalize homophily of network contacts, I counted the number of times respondents cited same-sex and opposite-sex actors in response to each of the five network questions. Two different homophily metrics were then derived for each respondent. The first indicates the number of cites to same-sex others as a proportion of total cites, for each of the five networks. For example, if a woman cited a total of six individuals, four men and two

1
I computed correlations between closeness and aggregate prominence centrality indices, which ranged from .98 to .99.

2
Marsden (1990) provided a comprehensive review of relevant factors in the determination of whether low reciprocation rates should be attributed to measurement error or, instead, to well-documented forms of systematic distortions.

women, as her advice contacts, her homophily on the advice network is .33. While this index is easily interpretable, it is biased in that it does not take into account the total availability of men and women, which precludes or makes possible any given choice pattern (Blau, 1977).

The second homophily metric corrected for this bias by calculating the following values for each respondent: (a) the number of ties a person sent to people of the same sex, (b) the number of ties a person sent to people of the opposite sex, (c) the number of people of the same sex the actor could have cited but did not, and (d) the number of people of the opposite sex the actor could have cited but did not. Homophily measures for each network (HComm, HAdvice, HSupport, HInfluence, HFriend) were then derived by the following calculation, which adjusts for both the availability of different-sized gender groups and individuals' choices:[3]

$$H = \sqrt{}[(a/a + b) - (c/c + d)]\,[(a/a + c) - (b/b + c)].$$

This calculation produces a measure ranging from -1 to 1; positive values indicate a tendency for a person to choose people of the same sex, given the availability of members of the same sex; a value of zero indicates a balanced mix of male and female choices, again, given availability. Respondents who chose no one in response to a given network question were assigned missing values, since there was no evidence to indicate how they would choose if they were to do so.

Multiplexity. A measure of multiplexity was obtained by counting, for each respondent or ego, the number of networks, out of a possible five, in which the same alter was chosen. The total number of uniplex ties was multiplied by one, the number of biplex ties by two, and so on, until ties to alters across all five networks were multiplied by five. Only the highest number of ties to alters was counted [i.e., such that an n-plex tie was not counted as an $(n - 1)$-plex tie, etc.]. A weighted average was produced by adding these values and dividing by the total number of ties possessed by the focal person, yielding a total average multiplexity across all possible relations to alters. The same procedure was used to obtain measures of multiplexity of ties to same-sex network contacts.

Individual characteristics and organizational positions. Sex is a dummy variable with a value of one assigned to males. Control variables included individual characteristics, formal organizational group memberships, and other work-related variables previously reported in the literature as factors associated with network integration. Education was coded as a three-level variable (0 = no college degree; 1 = college degree; 2 = graduate degree). Tenure in the organization was measured in years. Professional activity was coded as a three-level variable (0 = no professional activity; 1 = belongs to professional societies or attends meetings of professional societies; 2 = both belongs and attends). Prestige of past work experience was also coded as a dummy variable, a value of one indicating previous experience in Fortune 500 or Advertising 100 firms. This variable was included as a control because in the research site, a small advertising firm located away from major

3
Krackhardt (1990) has used this statistic (reviewed in detail by Gower and Legendre, 1986) with similar choice data. The measure, called S_{14}, has also been called the point correlation coefficient and is equal to the value obtained by computing a simple Pearson correlation between the two sets of dyadic observations (Krackhardt, 1990).

advertising centers, prior experience in a name-brand firm afforded employees considerable prestige.

Since the organization was composed of nine small departments, each respondent was assigned a value corresponding to membership in one of three larger groups described above (client management, core, and support). DEPT3 includes the Public Relations and Account Services departments; DEPT2 includes the Creative, Research, Production, and Traffic departments; and DEPT1 comprises the Media, Personnel, and Accounting departments. Department was thus coded as two dummy variables: DEPT2 and DEPT3, with DEPT1 as the reference category. Respondents were also assigned a score for each of the five designated hierarchical ranks (5 = senior management; 4 = mid-management; 3 = nonsupervisory professionals; 2 = entry-level employees; 1 = secretarial staff).

RESULTS

Variable means, standard deviations, and intercorrelations are reported in Table 2. The descriptive statistics and intercorrelations provided some measure of support for the theoretical distinction between instrumental and expressive networks. The friendship network appears to be most distinct from the others, reflecting the fact that it was the only network that is voluntarily chosen and taps interaction outside the domain of work. Although influence centrality was highly correlated with the other centrality indices, its highly skewed distribution (only 12 respondents, all but one of the senior managers and three middle managers, received a nonzero score) suggests that it is highly determined by formal authority. As might be expected, the lowest organizational mean centrality values were for the influence and advice networks (.10 and .25), while the highest means, indicating broader access, were obtained for the friendship and support networks (.42 and .45). Among the centrality indices, the lowest correlation obtained was between friendship and influence centrality (r = .35), followed by the correlation between advice and friendship centrality (r = .50). Advice, communication, and support centrality were the most highly correlated indices (correlations ranged from .76 to .86), followed by influence and advice (r = .73) and support and friendship (r = .65). These data suggest that the five networks may be viewed as points on a continuum of instrumental to expressive relations.

Sex Differences in Homophily, Multiplexity, and Centrality

Table 3 reports group differences in homophily of network ties. Means for both availability-adjusted and simple homophily scores (proportion of same-sex ties to total ties, reported in parentheses) are reported, but statistical analyses were only performed on the adjusted scores, the simple scores being merely illustrative.

T-test results suggest that, across the five networks, men send a significantly higher proportion of ties to men than women send ties to women. This pattern was most marked with respect to the influence and advice (instrumental) networks, where women, on average, sent a greater

Table 2

Means, Standard Deviations, and Correlations among All Variables (N = 79)

Variables	Means	S.D.	1	2	3	4	5	6	7	8	9	10	11	12	13
Centrality															
1. Communication	.38	.31													
2. Advice	.25	.31	.86$^{\bullet\bullet\bullet}$												
3. Support	.45	.25	.84$^{\bullet\bullet\bullet}$	.76$^{\bullet\bullet\bullet}$											
4. Influence	.10	.25	.64$^{\bullet\bullet\bullet}$	.73$^{\bullet\bullet\bullet}$	.59$^{\bullet\bullet\bullet}$										
5. Friendship	.42	.33	.62$^{\bullet\bullet\bullet}$	.50$^{\bullet\bullet\bullet}$	.65$^{\bullet\bullet\bullet}$	.35$^{\bullet\bullet\bullet}$									
Attributes															
6. Gender	.43	.50	.47$^{\bullet\bullet\bullet}$	.57$^{\bullet\bullet\bullet}$	.47$^{\bullet\bullet\bullet}$	.37$^{\bullet\bullet\bullet}$	.36$^{\bullet\bullet}$								
7. Prestige	.23	.42	.33$^{\bullet\bullet}$	.34$^{\bullet\bullet\bullet}$	.34$^{\bullet\bullet\bullet}$	.28$^{\bullet\bullet}$	.26$^{\bullet\bullet}$	.44$^{\bullet\bullet\bullet}$							
8. Professional activity	.62	.79	.47$^{\bullet\bullet\bullet}$	.44$^{\bullet\bullet\bullet}$	.37$^{\bullet\bullet\bullet}$	.52$^{\bullet\bullet\bullet}$	.24$^{\bullet}$	.16	.10						
9. Tenure	4.85	4.32	.36$^{\bullet\bullet}$	.34$^{\bullet\bullet}$	.41$^{\bullet\bullet\bullet}$	.54$^{\bullet\bullet\bullet}$	.10	.16	.05	.22$^{\bullet}$					
10. Education	1.98	.61	.24$^{\bullet}$	.28$^{\bullet}$	.35$^{\bullet\bullet}$	.20	.37$^{\bullet\bullet\bullet}$	.28$^{\bullet}$	.15	.07	.01				
11. DEPT1	.32	.47	−.37$^{\bullet\bullet\bullet}$	−.23$^{\bullet}$	−.34$^{\bullet\bullet\bullet}$	−.10	−.35$^{\bullet\bullet\bullet}$	−.32$^{\bullet\bullet}$	−.05	−.30$^{\bullet\bullet}$	−.20$^{\bullet}$	−.18			
12. DEPT2	.30	.46	.11	.14	.07	.06	.11	.26$^{\bullet\bullet}$	.03	−.21$^{\bullet}$	.27$^{\bullet\bullet}$	−.01	−.45$^{\bullet\bullet\bullet}$		
13. DEPT3	.38	.49	.25$^{\bullet}$	.09	.27$^{\bullet\bullet}$	.04	.23$^{\bullet}$	.06	.01	.48$^{\bullet\bullet\bullet}$	−.06	.18	−.53$^{\bullet\bullet\bullet}$	−.52$^{\bullet\bullet\bullet}$	
14. Rank	3.09	1.29	.75$^{\bullet\bullet\bullet}$	.77$^{\bullet\bullet\bullet}$	.66$^{\bullet\bullet\bullet}$	.58$^{\bullet\bullet\bullet}$	.38$^{\bullet\bullet\bullet}$	.55$^{\bullet\bullet\bullet}$	.34$^{\bullet\bullet\bullet}$	.30$^{\bullet\bullet}$	.29$^{\bullet\bullet}$	.26$^{\bullet}$	−.11	−.10	−.12

$^{\bullet}p < .05$; $^{\bullet\bullet}p < .01$; $^{\bullet\bullet\bullet}p \leq .001$.

proportion of ties to men (67 and 58 percent, respectively). The difference between men and women in the proportion of ties sent to same-sex others in the friendship network, where preferences for homophily were strong in both groups (76 and 73 percent of all ties to the same sex), was slight but significant when adjusted for availability. Thus, while men evidenced a great degree of sex-based homophily in their nominations across the five networks, women tended to show a more differentiated choice pattern: On average, they nominated a greater proportion of men as advice and influence ties, nominated men and women in near-equal proportions as sources of communication and support, and overwhelmingly nominated other women as friends.

OLS regression models with the homophily indices as dependent variables were used to test whether sex differences persist, holding individual attributes and formal positions constant. Unstandardized regression coefficients for gender were highly significant in all five models, with the coefficients for the support and friendship centrality models

Table 3

Differences between Men and Women in Homophily

Homophily indices	Means* Women	(N = 45)	Men	(N = 34)	T-value	d.f.	Unstandardized regression coefficients (N = 79)†
HComm	−.03	**(.49)**	.21	**(.78)**	−6.90$^{\bullet\bullet\bullet}$	77	.39$^{\bullet\bullet\bullet}$ (.06)
HAdvice	−.05	**(.42)**	.22	**(.86)**	−8.43$^{\bullet\bullet\bullet}$	77	.40$^{\bullet\bullet\bullet}$ (.05)
HSupport	−.003	**(.55)**	.17	**(.71)**	−5.65$^{\bullet\bullet\bullet}$	76	.19$^{\bullet\bullet\bullet}$ (.05)
HInfluence	−.09	**(.33)**	.22	**(.90)**	−10.90$^{\bullet\bullet\bullet}$	73	.39$^{\bullet\bullet\bullet}$ (.04)
HFriend	.06	**(.76)**	.14	**(.73)**	−2.57$^{\bullet}$	58	.17$^{\bullet\bullet\bullet}$ (.04)

$^{\bullet}p < .05$; $^{\bullet\bullet}p < .01$; $^{\bullet\bullet\bullet}p \leq .001$.

* Values in bold in parentheses are cites to same-sex others, as a proportion of total ties, unadjusted for availability.

† Unstandardized regression coefficients are reported for gender with homophily indices as dependent variables. All control variables (i.e., education, tenure, professional activity, prestige, department, rank) are included in the regression models; standard errors are in parentheses.

(expressive networks) evidencing a slightly lower level of significance. Overall, results reported in Table 3 provide strong evidence of homophily of network ties for men and a more differentiated pattern of relationships for women, providing support for hypothesis 1.

T-tests were also used to analyze differences between men and women in multiplexity of network ties. Members of both groups had, on average, approximately two types of network relationships with their contacts (2.23 for men versus 2.15 for women). On average, however, men reported a significantly greater average number of multiplex ties to men (2.42) than women reported multiplex ties to women (2.02). These differences remained significant despite controls, as indicated by the unstandardized regression coefficients reported on the right hand side of Table 4, providing partial support for hypothesis 2.

Table 4

Differences between Men and Women in Multiplexity

Multiplexity indices	Means: Women ($N = 45$)	Means: Men ($N = 34$)	T-value d.f. = 70	Unstandardized regression coefficients ($N = 79$)*
Total multiplexity	2.15	2.23	−.63	.19 (.14)
Multiplexity of ties to same sex	2.02	2.42	−2.42	.64•• (.21)

• $p < .05$; •• $p < .01$.

* Unstandardized coefficients are reported for gender with multiplexity indices as dependent variables. All control variables (i.e., education, tenure, professional activity, prestige, department, rank) are included in the regression model; standard errors are in parentheses.

Hypothesis 2, however, was not fully supported: Men and women did not differ significantly in their overall number of multiplex relationships. This hypothesis was based on the expectation that all homophilous ties are more likely to be multiplex. The relationship between homophily and multiplexity, however, differed by sex. For women, average homophily (across the five networks) was highly correlated with multiplexity of ties to women ($r = .71$, $p \leq .0001$) and total multiplexity ($r = .52$, $p \leq .0001$), while for men, homophily was independent of both total multiplexity and multiplexity of ties to men ($r = .19$ and $r = .13$, n.s., respectively). Thus, while women's strong (multiplex) ties tended to be homophilous, homophily was not correlated with multiplexity for men.

Table 5 reports differences in mean centrality scores for men and women. As expected, men had significantly higher centrality scores, as well as raw numbers of sociometric choices, than women across the five networks. Differences were most marked with respect to the advice network and least marked in the friendship network. Regression models, using the centrality scores as dependent variables, however, indicate that the effect of sex on centrality is entirely mediated by differences in background characteristics, rank, and department (the regression coefficient for gender in the

Table 5

Differences between Men and Women in Centrality

Centrality indices	Means* Women (N = 45)	Men (N = 34)	T-value d.f. = 70	Unstandardized regression coefficients (N = 79)†
Communication	.26	.55	−4.63•••	.04
	(3.9)	(8.7)		(.06)
Advice	.10	.45	−6.11•••	.08
	(2.0)	(6.8)		(.06)
Support	.35	.59	−4.69•••	.01
	(3.9)	(7.5)		(.05)
Influence	.02	.21	−3.53•••	.03
	(1.2)	(6.2)		(.05)
Friendship	.32	.56	−3.38•••	.02
	(2.3)	(3.9)		(.09)

• $p < .05$; •• $p < .01$; ••• $p \le .001$.

* Degree centrality scores, i.e., the raw number of nominations, are reported in parentheses.

† Unstandardized regression coefficients are reported for gender with centrality indices as dependent variables. All control variables (i.e., education, tenure, professional activity, prestige, department, rank) are included in the regression model; standard errors are in parentheses.

advice centrality model, however, approached significance at $p < .15$).

Sex Differences in Determinants of Centrality

Two sets of OLS regression analyses within gender-homogeneous groups, reported in Table 6, were performed to test the hypothesis that men are better able to convert valued resources into greater centrality (hypothesis 5). Analyses of determinants of influence centrality are not reported due to a severe restriction of range: only one of the 45 women in the organization had a nonzero centrality score on this network. The first set of regressions explores the unique contribution of attributes and formal positions to centrality in each network, for men and women separately. The second set investigates whether the addition of homophily and multiplexity makes a significant contribution to the percentage of variance explained, thus providing a direct empirical test of the effects of homophily on women's access to these networks.

Rank was the most significant predictor of communication, advice, and support centrality for both men and women, accounting for most of the variance explained, but the increments to centrality appeared to be greater for men, a finding consistent with previous reports (Miller, Lincoln, and Olson, 1981; Olson and Miller, 1983). Professional activity was a significant predictor of men's (but not women's) centrality in the communication and advice networks, although there were no significant differences between men and women on this dimension [$t(77) = 1.42$, n.s.]. Similarly, education contributed to men's friendship network centrality but not to women's.

Further, control of critical contingencies (membership in DEPT2 and DEPT3), appears to provide greater network returns to men. Within the friendship network, for example, men in these two departments obtained increments to centrality of .43 and .33 ($p < .05$), respectively, over men in

Table 6

Regression of Centrality on Independent Variables within Sex Groups*

Independent variables	Centrality Indices: Women (N = 45)				Men (N = 34)			
	Communication	Advice	Support	Friend	Communication	Advice	Support	Friend
Professional activity	.05	.07	−.01	.02	.11••	.13••	.03	.11
	(.05)	(.04)	(.04)	(.08)	(.05)	(.06)	(.05)	(.09)
Prestige	−.01	.03	.09	.20	.09	.02	.06	.08
	(.12)	(.10)	(.10)	(.19)	(.06)	(.07)	(.06)	(.11)
Tenure	.004	.01	.02•	.003	.01	−.01	.001	−.02
	(.01)	(.01)	(.01)	(.01)	(.01)	(.01)	(.01)	(.01)
Education	−.00	.03	.07•	.09	.03	.02	.01	.22••
	(.04)	(.04)	(.04)	(.07)	(.06)	(.07)	(.06)	(.10)
DEPT2	.17••	.03	.05	.09	.27•••	.21•	.12	.43••
	(.09)	(.07)	(.08)	(.14)	(.10)	(.11)	(.10)	(.17)
DEPT3	.14	.01	.14•	.15	.20••	−.01	.12	.33••
	(.08)	(.07)	(.08)	(.14)	(.09)	(.10)	(.10)	(.16)
Rank	.14••••	.11••••	.08•••	.05	.24••••	.26••••	.16•••	.11
	(.03)	(.02)	(.02)	(.04)	(.04)	(.05)	(.05)	(.08)
R^2	.55••••	.48••••	.47••••	.19	.76••••	.72••••	.59••••	.36•
Adjusted R^2	.46	.37	.37	.03	.69	.64	.49	.19
AVHomophily	−1.10•	−.62	−1.22••	−1.87•	.76••	.52	.38	.74
	(.50)	(.45)	(.43)	(.81)	(.35)	(.43)	(.38)	(.68)
Multiplexity to same sex	.06	.05	.04	.15	−.07	−.07	−.02	−.03
	(.05)	(.05)	(.04)	(.08)	(.06)	(.07)	(.06)	(.11)
R^2	.61••••	.51••	.60••••	.30	.80••••	.74••••	.61••	.39•
ΔR^2	.06•	.03	.13••	.11•	.04•	.02	.02	.03
Adjusted R^2	.50	.37	.49	.11	.73	.65	.47	.17

•$p < .10$; ••$p < .05$; •••$p < .01$; ••••$p \leq .001$.
* Unstandardized regression coefficients; standard errors are in parentheses.

low-status DEPT1; for women, however, the incremental contribution to centrality of membership in these two departments was .09 and .15 (n.s.), respectively. A similar if more modest effect is evidenced in the male and female regression coefficients for DEPT2 in the advice centrality model and for DEPT2 and 3 in the communication centrality models.

Overall, the results reported in the upper portion of Table 6 are in the expected direction: On no dimension did women appear to reap greater network rewards than men. One exception is the positive contribution of tenure to women's support network centrality, while the significant association between tenure and men's communication, advice, and support centrality (not reported: .45, .41, and .46, $p <$ .01–.05) disappeared once controls were instituted. Thus, greater tenure without greater rank is of no advantage to men but provides greater centrality to women. This is consistent with results reported by Olson and Miller (1983), who argued that men are penalized to a greater extent than women for a lack of mobility.

The bottom portion of Table 6 reports the results of including average homophily and multiplexity of ties to same-sex others in the models. Average homophily appears to detract significantly from women's communication, support, and friendship centrality. The addition of these variables produces a significant ($p < .01$) change in R-squared in women's support network centrality, as well as near significant ($p < .10$) changes in communication and friendship network centrality. Results were particularly striking in the case of support network centrality, where only rank approximated the contribution of homophily. By contrast, homophily contributes positively to men's communication centrality, producing a near significant ($p < .10$) change in R-squared.

In order to ascertain the extent to which the above results support the hypothesis of differential returns to men and women, tests of the equality of regression slopes across groups were performed. Table 7 reports the significant interaction terms obtained in regression models that included all control variables. The interaction terms (gender* control variable) were entered into the regressions one at a time, and only significant ($p < .10$) terms were included in the reported models. The most substantial differences of slopes were those pertaining to the contribution of homophily to support and friendship network centrality, although rank and professional activity appeared to yield differential returns to advice centrality for men and women. Again, the distinction between instrumental and expressive networks appears to be useful: Bureaucratic investments (i.e., rank and professional activity) provide lesser returns to women in a key instrumental network (advice), while homophily primarily detracts from their centrality in expressive networks. Overall results are consistent with hypothesis 5, asserting that men are better able to convert valued resources into network centrality.

Table 7

Tests of Equality of Regression Slopes across Sex Groups (N = 79)*

	Dependent Variables			
Interaction terms	Communication	Advice	Support	Friendship
Sex * Professional activity	–	.03•• (.06)	–	–
Sex * Rank	–	.03• (.02)	–	–
Sex * AVHomophily	.85• (.48)	–	1.19••• (.43)	1.68•• (.79)
ΔR^2	.014•	.02•	.04•••	.04••

• $p < .10$; •• $p < .05$; ••• $p < .01$.
* Unstandardized regression coefficients; standard errors are in parentheses. Control variables are the same as in Table 6.

DISCUSSION AND CONCLUSION

The analyses reported above confirm the major propositions of this study. Men and women differed in homophily and multiplexity of network ties and in determinants of network centrality. The pattern that emerges, however, when examined in light of contrasting theoretical perspectives, is not the simple picture of female exclusion portrayed in much previous research.

Sex Differences in Homophily: Choice and Structural Constraint

Despite controls for availability, men evidenced a greater degree of homophily across the five networks, while women tended to show a differentiated pattern of tie choice, with greater heterophily in their instrumental than expressive networks. That is not to say that men were not selective in their choice of network contacts: The lack of correlation between homophily and multiplexity for men indicates that they choose different men for different types of network relationships. The "division of labor," however, is within rather than across gender groups. A critical question is whether these homophily findings should be attributed to male exclusion of women (constraints that are independent of availability) or to women's strategies (choice homophily) for obtaining access to the best of two worlds—social familiarity and instrumental access. The two may be difficult to untangle.

While the male homophily results may be viewed as evidence of constraint on women's choices, the argument can also be made that both men and women appear to be making rational choices in favor of higher-status instrumental contacts. But the two sexes do not exhibit equivalent "preferences" for homophily or status. While these analyses did not allow me to distinguish between preferences for similarity and status in men's choices, it is clear that women did not always direct the majority of their ties to men (because they are more powerful) or to women (because they are more similar) but, instead, differentiated their choices, plausibly so as to obtain greater access to both expressive and instrumental resources. The dilemma for women, however, may lie in men's reciprocation of their choices: If network contacts are chosen according to similarity and/or status considerations, they are less desirable network choices for men on both counts. And that highly significant differences in homophily persist despite controls for human capital and formal position indicates that differences in choice patterns are not attributable to differences in the formal positions of the focal (i.e., choosing rather than chosen) individuals.

As might be expected, given the distribution of men and women across departments and hierarchical levels, homophily contributed to or had no effect on men's centrality, while it decreased women's centrality. While this might be interpreted in light of the high correlation between homophily and multiplexity for women (and not for men), homophily detracted from centrality in the presence of controls for multiplexity. The critical issue thus may not be whether or not women's networks constitute dense webs of strong ties (Aldrich, 1989) but simply that the network resources reached through ties to women are relatively poor, regardless of the strength of the ties.

Of additional interest is the finding that average homophily across the multiple networks, rather than homophily in a given network, was responsible for reduced centrality in that same network (e.g., analyses not reported in the paper indicate that homophily in the friendship network was not a significant predictor of centrality in the friendship network, and so on for each of the networks). Thus, what may appear to be optimizing choices for women with respect to one-on-one relationships (i.e., choosing female expressive contacts to maximize social support) may produce suboptimal effects, since average homophily appears to reduce women's centrality within the very same expressive networks. Further, the finding that homophily has more detrimental effects on expressive rather than instrumental centrality appears at first blush counterintuitive. This may be explained partially by the finding that men in this firm, while predominantly choosing network contact with men, appeared to include more women in their friendship and support networks than in their instrumental choices. To the extent that women, by contrast, made their most homophilous choices in this domain, they did not fully take advantage of a potential "opening" by the dominant group.

Do these findings, then, support Brass's (1985: 340) conclusion that "encouraging women to form networks with other women in the same organization may be unnecessary, or, at worst, nonproductive"? An unequivocal answer cannot be provided by the data reported here. Results, however, are consistent with Kanter's (1977: 282) assertion that women's networks are "more effective when they are task-related and have a meaningful function to perform for the organization, instead of more peripheral social clubs (which reinforce stereotypes about women's greater interest in talk than tasks)." More broadly, this research does suggest that women are likely to benefit from the development of greater ties to their male colleagues; whether they were unable or unwilling to do so in this organization cannot be answered by the data.

Sex Differences in Centrality: Systemic Barriers and Human Capital

The notion that women are denied access to male-dominated interaction networks regardless of bureaucratic investments was also not fully supported in the analyses of network centrality. When analysis of sex differences was limited to overall level of centrality, the human capital hypothesis was supported: The effect of gender on centrality was entirely mediated by differences in background characteristics, rank, and department. Analyses of sex differences in determinants of centrality, however, lent support to the systemic-barriers notion that when the workforce is highly differentiated by ascribed attributes such as gender, these attributes take on added significance in organizational processes (Miller, 1986). In particular, rank and professional activity appeared to provide differential returns to men and women.

The question that remains and cannot be fully answered within the context of this research is what accounts for gender differences in the ability to convert human capital investments and positional resources into network centrality. Speculative hypotheses may be derived from Spence's (1974) notion of "market signaling," in which he argues that human capital influences job and promotion decisions only to the extent that these assets can be observed by decision makers. As noted by Barney and Lawrence (1989), however, many of the assets of interest to decisionmakers are not easy to observe accurately.

Extending this notion to the ability to convert human capital and formal positional resources into network access, it is possible that while human capital and positional resources may "signal" attractiveness as a network contact for men, for women, network access itself may be the signal that the resources they possess are of value. This speculation is consistent with recent findings that suggest that women need strong network relationships with strategic partners in order to provide evidence or "cues" of their legitimacy as key players (Burt, 1992). Support for a signaling hypothesis may also be inferred from research suggesting that male mentors are less likely to assume that women are competent and thus defer establishing relationships with women protégés until they have proven themselves (Newton and Fitt, 1981). As Keele (1986) noted, those most likely to be attractive as protégés for a mentor are those who demonstrate that other contacts are available to them. Extrapolating to the current context, women may be more likely than men to be caught in the Catch-22 of needing network relationships in order to be able to convert fully their individual and positional resources into network access.

Implications for Research

These findings have multiple implications for organizational theory and research. First, study results suggest that the distinction between induced and choice homophily (McPherson and Smith-Lovin, 1987), while highly useful, should also consider constraints that cannot be attributed to strict numerical availability and preferences that may be based on factors other than similarity. Further, given that sex is not the only basis for homophily, further research is needed to investigate whether sex-based homophily is a more important basis for perceived similarity than, for example, sharing a similar educational background or occupation, as well as to explore potential interaction effects among varying forms of homophily (Brass, 1985). By the same token, more fine-grained analyses that would allow us to unravel or distinguish between similarity and status considerations in nomination patterns are needed to shed light on some of the questions raised by this study.

A second area of research on which the present findings have bearing is the advantages and disadvantages of weak and strong network ties (Granovetter, 1973, 1982). While tie strength was not measured directly in this study (traditional indicators include intimacy of the relationship and frequency of interaction), that women tended to gain instrumental access and expressive support from different sources implies a prevalence of weak ties that may preclude or inhibit the development of strong mentoring relationships (Kram, 1985). Research is needed to explore the conditions under which strong and weak ties provide network benefits or disadvantages to women and to investigate optimal combinations of each (Brass, 1992). Similar issues are likely to come into play for racial minorities, since network contact with both minorities and majority group members is required to obtain access to social support and instrumental resources (Bell, 1990; Thomas, 1990).

The need for further research on the strength of ties is especially acute in light of two seemingly contradictory perspectives currently offered in the literature. Lin (1982) argued that weak ties, because they are more likely to connect people who differ on status, are the only available avenues for access to resources for low-status individuals. The negative effects of homophily on women's centrality are consistent with this view. A second body of work, however, suggests that weak ties are less likely to be useful to individuals in "insecure positions" or lacking in legitimacy (Granovetter, 1982; Burt, 1992). Burt (1992), for example, found weak-tie-based networks to be more advantageous from a mobility standpoint for men than for women, who required strong ties to strategic partners to signal their legitimacy and thus contribute to their advancement. This alternate view also seems to be consistent with the notion of signalling discussed above. Burt (1992) also reported, however, that strategic partners were only useful to women to the extent that their networks were rich in weak ties. Thus, for women or members of other groups lacking in legitimacy, weak ties may be critical—although those that are most useful may not be their own. The key issue, therefore, may not simply be an appropriate balance of strong and weak ties, but who they link and, thus, the level and types of resources available through each.

Finally, the speculative interpretations provided here concerning signalling effects merit further investigation. While men's greater ability to convert human capital into positional resources (high-status jobs, higher rank and salary) has been firmly established (Cole, 1979; Baron and Bielby, 1985), theories explaining men's advantage in converting positional resources into network access have yet to be developed. Research to investigate whether different signalling processes are in operation for men and women seems promising. Alternately, further research is needed to investigate whether or not similar network patterns provide the cues that translate into similar advancement outcomes.

Limitations

The most important methodological limitation of this study is common to most network studies (see Barley, 1990, and Burkhardt and Brass, 1990, for notable exceptions): In cross-sectional research, causal relationships are difficult to untangle. Rather than homophily producing greater or lesser centrality, it is possible that women who experience exclusion from organizational networks are more likely to seek refuge in ties to other women. Similarly, the causal relationship between hierarchical position and network centrality is ambiguous, with centrality potentially preceding promotion to a higher rank (Brass, 1985). Moreover, cross-sectional designs obscure dynamic processes; research that examines the processes and behavioral strategies by which network bonds are developed and used is needed to further the development of theoretical explanations for sex differences. Network differences may be the result of men and women using different approaches to achieve network access or, instead, using a similar approach but with less skill or less success or both (Ragins and Sundstrom, 1989).

A second potential limitation stems from the fact that the entire secretarial staff was not included in the study population. One might argue that, had they been included, average homophily for women might have been higher, potentially reducing the significant difference between men and women on this dimension. It is also reasonable to assume, however, that any difference strictly attributable to position on the lowest rung of the organization would disappear in the presence of controls for hierarchical position.

Finally, the specific demographic configuration or unique set of circumstances of the organization studied may limit the generalizability of these results to other settings. The high level of homophily observed may be partially explained by the environmental turbulence the firm experienced in the years prior to data collection. Under turbulent task environments and conditions of uncertainty, individuals assume a more defensive posture and are more likely to direct their networking efforts to others with a common background (Galaskiewicz and Shatin, 1981). In addition, the firm tended to exclude women from high-status jobs and certain hierarchical levels; whether the same approach in an organization with a different demography would yield the same results remains an empirical question (Brass, 1985). Yet Baron and Bielby (1985: 236) reported that "most organizations are either perfectly segregated or closely approximate that state of affairs," and findings of differential network returns have been obtained in contexts characterized by greater balance (Miller, Lincoln, and Olson, 1981; Olson and Miller, 1983; Miller, 1986). Further, South et al. (1982) found that a high level of female representation in workgroups diminishes the amount of social support that women receive from male coworkers, providing support for the alternative hypothesis that the larger a minority group, the greater the discrimination against it.

While the fine details of the findings may not generalize far beyond the current research setting, the more general interpretations and implications of the findings are likely to have wider applicability, particularly to other professional service firms facing similar critical contingencies (i.e., client dependence) and characterized by similar levels of female exclusion from high-status positions. The study's findings are generally consistent with existing network research, suggesting that they are applicable to small firms, which, as Lincoln (1982: 14) noted, "rely less on formal mechanisms of control and stress personal face-to-face processes of social influence and coordination." The relative dearth of network research in large industrial corporations, however, raises more serious questions about the generalizability of these findings to such a setting.

This paper has argued that sex differences in homophily and in rates of return to individual and positional resources operate to create and reinforce gender inequalities in the organizational distribution of power. The study also highlights the importance of investigating the relationship between homophily and centrality as it affects network inequalities. Few network studies to date have incorporated both ego-network variables—properties of relations that a focal actor has with others in his or her personal network such as homophily—and variables such as centrality, which reflect relations that a focal individual has with the entire population of actors in a given system, including those to whom he or she is not linked. These results suggest that studying the properties of ego networks may elucidate larger structural patterns and their consequences, and they indicate the value of incorporating different levels of analysis and different network types in future studies.

Network analysis offers a rigorous methodological and theoretical framework for the study of social influences that perpetuate disparities in organizational experiences. Further research can assess whether actual or perceived differences are impeding women's progress in the workplace and shed light on the extent to which informal networks contribute to differences in the distribution of power for men and women.

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